

RAMAKRISHNA NAKKA

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with hands-on experience building production AI applications — multi-LLM orchestration pipelines, real-time voice assistants, and serverless browser extensions. Comfortable working end-to-end across Python backends and React frontends. Approaches AI-assisted development with a specification-first discipline: authoring clear feature documents before engaging AI tooling, using agentic IDEs daily, and holding four Anthropic certifications across the Claude API, Claude Code, Agent Skills, and MCP.

PROFESSIONAL EXPERIENCE

DeepKlarity — Software Engineer *Jan 2026 – Present*

AI/IoT Platform · Python · FastAPI · React · LangChain · Multi-LLM Orchestration

- **GROK AI Engine:** Contributed across the full stack of an internal AI platform that converts natural-language IoT/AIoT problem statements into executable Blockly XML via a two-stage LLM pipeline (NL → Pseudocode → Code).
- **Multi-Provider LLM Integration:** Extended the orchestration layer to support Google Gemini 2.5 Pro alongside Claude Sonnet and GPT-4o — updated LangChain manager, dependency config, and model-selection UI across both chat interfaces.
- **Generation Health Dashboard:** Designed and built a full-stack SuperAdmin analytics dashboard (FastAPI REST + React) that tracks pipeline success and failure rates by prompt and model, enabling data-driven LLM prompt optimization.
- **Debug Tooling & Bug Resolution:** Overhauled primary debug tooling with responsive collapsible panels and restructured API routes; resolved 8 production bugs spanning state synchronization, sensor-name normalization, and multi-environment routing.

DeepKlarity — Full-Stack Developer Intern *Oct 2025 – Jan 2026*

Sarvam AI Integration · Python · FastAPI · React Native · WebSockets · Chrome Extensions

- **Smart Mandi — Real-Time Voice AI:** Sole engineer on a voice assistant enabling Indian farmers to query live commodity prices in their native language. Architected a WebSocket audio pipeline (mobile M4A/AAC → server-side PCM-16 via FFmpeg), a dynamic context injector that fetches Redis market data just-in-time before LLM inference, and Pipecat-based VAD orchestration across Sarvam AI speech, language, and TTS models.
- **Indian Language Subtitle Generator — Chrome Extension:** Built a serverless Manifest V3 extension that captures live YouTube audio via chrome.tabCapture, routes it through Sarvam AI's transcription and codemix-aware translation APIs, and injects synchronized subtitle overlays into the YouTube video DOM. Implemented per-video caching via chrome.storage.local to reduce API consumption on repeat views.

PROJECTS

Smart Mandi | *Python · FastAPI · WebSockets · Pipecat · Redis · FFmpeg · React Native · Sarvam AI*

- End-to-end real-time voice AI assistant for Indian farmers: streams audio over WebSockets with on-the-fly transcoding, injects live commodity prices via a custom MandiContextInjector module, and returns spoken responses using Sarvam AI's STT, LLM, and TTS models.
- Engineered a custom WebSocket serializer bridging mobile M4A/AAC audio to server-side PCM-16; Pipecat handles VAD and conversational turn-taking to ensure low-latency, natural interaction.

Indian Language Subtitle Generator | *React 18 · Vite · TailwindCSS · Chrome Manifest V3 · Sarvam AI (Saaras + Mayura)*

- Serverless Chrome Extension that captures live browser audio without a backend, processes it through Sarvam AI with codemix support (Hinglish/Tanglish), and injects subtitles synchronized to the YouTube HTML5 player timeupdate event.
- Implemented edge caching via chrome.storage.local indexed by video ID and target language, reducing repeat API calls and improving load times on previously viewed content.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Backend: FastAPI, Node.js, SQLAlchemy (async), Pydantic, LangChain, WebSockets, REST APIs

Frontend: React 18, React Native, Vite, TailwindCSS, React Router, Axios

AI / LLM: LangChain, Langfuse, Pipecat, Anthropic API, OpenAI API, Google Gemini API, Sarvam AI (STT/LLM/TTS), Prompt Engineering

Infra & Tools: PostgreSQL, Git, GitHub, JWT, RBAC, Nginx, Supervisor

Agentic IDEs: Cursor, Windsurf, GitHub Copilot, Qwen

EDUCATION

National Institute of Technology (NIT) Mizoram 2021 – 2025

Bachelor of Technology, Computer Science and Engineering

CERTIFICATIONS

Anthropic

Claude with the Anthropic API ([Link](#))

Claude Code in Action ([Link](#))

Introduction to Agent Skills

Introduction to Model Context Protocol ([Link](#))

NxtWave

Python Programming Foundations ([Link](#))

Developer Foundations ([Link](#))

ACHIEVEMENTS

- **DSA Competitive Programming:** Rank 3 — CodeVerse Bi-Weekly #22 ([Link](#)) Rank 4 — CodeVerse Bi-Weekly #20 ([Link](#))
- **GitHub Badges:** YOLO Pull Shark ×2 Pair Extraordinaire Quickdraw